

# Topic 05

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## Programming Fundamentals – Tutorial Sheet Iteration – Loops (For and While)

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### Part A: Understanding While Loops

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#### 1. Writing simple while loops

Write Java code for each of the following:

1. Print the message **"Hello World"** five times using a **while** loop.
  2. Print the numbers from **1 to 10** (inclusive) on separate lines.
  3. Print the numbers from **10 down to 1** (inclusive) on the same line, separated by spaces.
  4. Print the numbers from **5 to 12** (inclusive) on separate lines.
  5. Print the numbers from **20 down to 10** (inclusive) on the same line, separated by spaces.
  6. Print the numbers from **2 to 17** (inclusive) in steps of 3 on separate lines
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#### 2. Predicting output

Without running the code, write down exactly what is printed by the following program:

```
int i = 2;
while (i <= 8) {
    System.out.println("Value of i: " + i);
    i = i + 2;
}
```

- a. How many times does the loop run?
  - b. What is the last value of **i** printed?
- 

#### 3. Correcting logic

The following loop is supposed to print the numbers 1–5, but it doesn't work. Explain what's wrong and correct the code:

```
int i = 1;
while (i <= 5) {
    System.out.println(i);
}
```

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#### 4. Designing your own

Write a **while** loop that prints the **square of each number** from 1 to 5:

```
1
4
9
16
25
```

## Part B: For Loops

### 5. Rewriting while as for

Rewrite each of the following `while` loops using a `for` loop.

(a)

```
int i = 1;
while (i <= 4) {
    System.out.println("Java");
    i++;
}
```

(b)

```
int n = 10;
while (n >= 1) {
    System.out.println(n);
    n--;
}
```

### 6. Writing simple for loops (rewriting the while loops from Part A)

Write Java code for each of the following:

1. Print the message **"Hello World"** five times using a `for` loop.
2. Print the numbers from **1 to 10** (inclusive) on separate lines.
3. Print the numbers from **10 down to 1** (inclusive) on the same line, separated by spaces.
4. Print the numbers from **5 to 12** (inclusive) on separate lines.
5. Print the numbers from **20 down to 10** (inclusive) on the same line, separated by spaces.
6. Print the numbers from **2 to 17** (inclusive) in steps of 3 on separate lines

### 7. Tracing nested loops

What will this code print?

```
for (int i = 0; i < 2; i++) {  
    for (int j = 0; j < 3; j++) {  
        System.out.println("i = " + i + ", j = " + j);  
    }  
}
```

- How many total lines are printed?
  - What is printed on the 5th line?
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## 8. Writing a counting pattern

Write a `for` loop to print:

```
Counting up: 1 2 3 4 5  
Counting down: 5 4 3 2 1
```

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## 9. Thinking challenge

Why might a **while loop** be useful for **input validation** (for example, asking a user for a positive number)?

Write an example snippet to demonstrate your reasoning. Use the Loop Control Variable concept.

- What would your Loop Control Variable be?
- How would you initialise it?
- How would you update it?
- How would you test it?